

# AI Logistics Simulator User Manual

This manual lists every control and every output metric, with **allowed values / ranges, units**, and **how to read** them in the prototype.

## 1) Before You Run

1. Start the FastAPI backend: `uvicorn backend.app:app --reload`
2. Start the simulator UI: `streamlit run backend/streamlit_control_tower.py`
3. Open the Streamlit URL from the terminal (commonly `http://localhost:8501`).

After you change backend logic, **restart the API** (or rely on `--reload`); the UI only needs a refresh if the page is stale.

## 2) Sidebar Controls

| Control                       | Range / values                                               | Default                            | Interpretation                                                                                                                                                                |
|-------------------------------|--------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FastAPI Base URL              | Any valid HTTP base URL for your API                         | <code>http://127.0.0.1:8000</code> | Where the UI sends <code>POST</code> requests. Must match the running <code>uvicorn</code> host and port.                                                                     |
| Mode                          | <code>sync</code> · <code>async</code> · <code>replay</code> | <code>sync</code>                  | <b>sync</b> : standard orchestration. <b>async</b> : same flow via <code>async</code> runtime. <b>replay</b> : same result plus a <b>per-node execution trace</b> (timeline). |
| View                          | Classic Simulator · 10 Challenges Demo                       | 10 Challenges Demo                 | <b>Classic</b> : summary KPIs + notes + replay trace. <b>10 Challenges</b> : one tab per use case (demand, route, ...) with detail.                                           |
| Show existing raw JSON output | On / Off                                                     | Off                                | <b>On</b> : shows raw <code>execution_meta</code> and <code>tool_outputs</code> for debugging. <b>Off</b> : business-friendly view only.                                      |
| Fetch Eval Metrics            | Button                                                       | —                                  | Fetches <code>GET /agents/eval-metrics</code> and displays offline calibration metrics.                                                                                       |
| Auto-refresh Eval Metrics     | On / Off                                                     | Off                                | Periodically refreshes eval metrics with a short cooldown after chat activity.                                                                                                |

## 3) Real-life Scenario Builder (before form)

Use this first to auto-generate realistic conditions from route context:

- **Collection Point (Start)**
- **Delivery Point (Destination)**
- **Scenario Vehicle Type** (van or motorcycle)
- **Create Scenario (Google Maps + Weather Forecast)**

What it does: - Computes distance/time/traffic and weather - Estimates `demand_index` + `fleet_health_index` - Auto-fills key simulation fields in the form

If Google key is missing, it uses fallback mode and shows a notice.

## 4) Scenario Form (Inputs)

All of these are sent in the JSON body when you click **Run Simulation**. Sliders use **inclusive** min/max as shown in the UI.

| Field                            | Type           | Range / allowed values                                                                                    | Default              | Step              | Interpretation & effect in the prototype                                                                                                                                                                                                                 |
|----------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Region                           | Select         | <code>SG</code> · <code>MY</code> · <code>PH</code> · <code>ID</code> · <code>TH</code> · <code>VN</code> | <code>SG</code>      | —                 | Forecast context label and scenario market.                                                                                                                                                                                                              |
| Demand Index                     | Slider         | <code>0.60</code> - <code>1.80</code> (unitless index)                                                    | <code>1.15</code>    | <code>0.05</code> | <b>1.0 = baseline demand</b> . Below 1.0 reduces pricing pressure; above 1.0 increases it (see <b>Demand multiplier</b> under outputs). Also nudges <b>last-mile failure</b> slightly when > 1.0.                                                        |
| Weather                          | Select         | <code>clear</code> · <code>rain</code> · <code>storm</code>                                               | <code>clear</code>   | —                 | Affects <b>route ETA</b> (rain/storm = longer ETA) and <b>last-mile failure</b> (mapped to internal severity for the failure model).                                                                                                                     |
| Urgency Mix                      | Select         | <code>standard</code> · <code>express</code> · <code>same_day</code>                                      | <code>express</code> | —                 | Affects <b>dynamic price</b> (higher urgency → higher multiplier) and <b>carbon recommendation</b> text; nudges <b>last-mile failure</b> (tighter service = slightly higher risk in the prototype).                                                      |
| Fleet Health Index               | Slider         | <code>0.50</code> - <code>1.00</code> (0 = poor, 1 = excellent)                                           | <code>0.90</code>    | <code>0.01</code> | Drives <b>simulated prior failed delivery attempts</b> (lower health → more “prior attempts” in the failure model) and signals <b>maintenance</b> context in the story; maintenance tool still uses its own fixed demo inputs unless you extend the API. |
| Traffic Level                    | Select         | <code>low</code> · <code>medium</code> · <code>high</code>                                                | <code>medium</code>  | —                 | Affects <b>route ETA and fuel</b> (higher traffic → longer ETA, more fuel) and nudges <b>customer presence</b> in the <b>last-mile failure</b> model.                                                                                                    |
| Vehicle Type (for carbon saving) | Select         | <code>diesel</code> · <code>EV</code> · <code>hybrid</code>                                               | <code>diesel</code>  | —                 | Used for carbon/emissions optimization and recommendation.                                                                                                                                                                                               |
| Tool Retries (Advanced)          | Integer slider | <code>0</code> - <code>3</code>                                                                           | <code>1</code>       | <code>1</code>    | Reliability setting under <b>Advanced / Reliability Settings</b> .                                                                                                                                                                                       |
| Tool Timeout (s) (Advanced)      | Slider         | <code>1.0</code> - <code>10.0</code> seconds                                                              | <code>3.0</code>     | <code>0.5</code>  | Reliability setting under <b>Advanced / Reliability Settings</b> .                                                                                                                                                                                       |

5) How to Run a Scenario

- 1. Set **Mode**, **View**, and optional **raw JSON** in the sidebar.
- 2. Adjust the scenario form.
- 3. Click **Run Simulation**.
- 4. Read the **status** line, then the **metrics** for your selected view (see sections 5-6).

6) Top dashboard metrics (Plan + classic header)

These come from the orchestration **plan** and **execution\_meta** and appear in **Classic** view and/or the **10 Challenges** header row.

| Metric                                               | Typical range in prototype                                                      | Unit                             | Where it comes from                                                                                                                                                        | How to read it                                                                                                                                                                    |
|------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target ETA (min)                                     | About <b>tens of minutes</b> (depends on scenario distance x traffic x weather) | Minutes                          | <b>Route</b> tool: distance + traffic penalty x weather penalty → <code>estimated_eta_min</code>                                                                           | <b>Lower is faster</b> . Rises with <b>high traffic</b> and <b>rain/storm</b> .                                                                                                   |
| Price Quote                                          | ≥ <b>3.50</b> in the pricing heuristic (then scaled by urgency and demand)      | Currency units (unnamed in demo) | <b>Pricing</b> tool: base distance + weight + <b>urgency multiplier</b> x <b>demand multiplier</b> + fuel surcharge                                                        | <b>Higher</b> urgency / distance / demand → <b>higher</b> quote. <b>Demand multiplier</b> = <code>1 + (demand_index - 1) x 0.2</code> (so at index <b>1.0</b> it is <b>1.0</b> ). |
| Failure Probability                                  | <b>0.000 – 0.990</b> (capped)                                                   | Probability (0-1)                | <b>Last-mile / delivery failure</b> tool: weighted mix of <b>customer presence</b> , <b>address quality</b> , <b>weather severity</b> , <b>prior attempts</b> , <b>COD</b> | <b>Higher = more last-mile risk</b> in the toy model. Varies with <b>weather</b> , <b>traffic</b> , <b>fleet health</b> , <b>demand</b> , <b>urgency</b> (see section 3).         |
| Forecast Volume                                      | <b>Non-negative integer</b> (shipment count style)                              | Count / day (conceptual)         | <b>Forecast</b> tool: rolling history + promo/holiday flags (defaults are fixed in orchestration)                                                                          | <b>Higher</b> volume vs baseline tends to <b>higher</b> capacity risk.                                                                                                            |
| CO <sub>2</sub> Estimate (kg) (10 Challenges header) | ≥ <b>0</b>                                                                      | kg CO <sub>2</sub> (approximate) | Derived from carbon + route context with selected vehicle class ( <code>diesel</code> / <code>EV</code> / <code>hybrid</code> )                                            | <b>Lower is better</b> for the same trip; vehicle class has large impact.                                                                                                         |
| KPI Deltas                                           | Signed values                                                                   | mixed                            | <code>plan.kpi_deltas</code>                                                                                                                                               | Shows delta vs baseline for ETA, cost, risk, CO <sub>2</sub> .                                                                                                                    |
| Dispatch Buffer (%)                                  | <b>3, 8</b> , or <b>15</b> in the current forecast rules                        | Percent                          | <b>Forecast</b> <code>recommended_buffer_pct</code> from <b>capacity_risk</b>                                                                                              | <b>Higher</b> buffer when <b>capacity_risk</b> is <b>high</b> (suggested extra operational slack).                                                                                |
| Risk Review Triggered                                | <b>Yes / No</b>                                                                 | Boolean                          | <code>execution_meta.risk_review_triggered</code> (tied to whether "risk_review" appears in <b>agent notes</b> in the current code)                                        | <b>Yes</b> = risk path was noted in the trace; use <b>replay</b> + <b>raw JSON</b> to inspect.                                                                                    |
| Trace Steps (replay mode)                            | <b>Integer ≥ 1</b>                                                              | Count                            | <b>Replay</b> stream: one step per graph event                                                                                                                             | <b>Larger</b> = more node updates recorded in the trace.                                                                                                                          |
| Agent Notes                                          | List of short strings                                                           | —                                | Appended by each agent node in the graph                                                                                                                                   | <b>Narrative</b> of what the orchestration “did” in order.                                                                                                                        |
| Replay Timeline (replay mode)                        | Steps with <b>node name</b> , <b>elapsed ms</b> , <b>keys updated</b>           | Time (ms)                        | <b>Replay</b> trace                                                                                                                                                        | <b>Explainability</b> : which node ran and which state keys changed.                                                                                                              |

7) “10 Challenges Demo” tab metrics

Each tab shows tool output. Ranges below match the **prototype** math in the backend, not live production data.

Tab 1 - Demand (forecast)

| Metric               | Range / values                          | Unit  | Interpretation                                                                                                                            |
|----------------------|-----------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Predicted Volume     | Non-negative integer                    | Count | Model “next day” style volume from <b>historical</b> series + <b>promo/holiday</b> multipliers (defaults fixed unless scenario extended). |
| Trend                | <code>upward · stable · downward</code> | —     | Class vs recent baseline ( <b>±15%</b> / <b>–10%</b> style thresholds in code).                                                           |
| Capacity Risk        | <code>low · medium · high</code>        | —     | <b>high</b> if volume > baseline x <b>1.2</b> ; <b>medium</b> if above baseline; else <b>low</b> .                                        |
| Recommended buffer % | <b>3 / 8 / 15</b>                       | %     | Tied to <b>capacity_risk</b> ( <b>high</b> → <b>15</b> , <b>medium</b> → <b>8</b> , <b>low</b> → <b>3</b> ).                              |

Tab 2 - Route (optimize\_route)

| Metric        | Range / values                         | Unit    | Interpretation                                                                        |
|---------------|----------------------------------------|---------|---------------------------------------------------------------------------------------|
| ETA (min)     | Positive integer (prototype heuristic) | Minutes | Same as <b>Target ETA</b> drivers: traffic x weather x fixed distance/waypoints.      |
| Distance (km) | Positive decimal                       | km      | Uses scenario-generated route distance when available; otherwise heuristic distance.  |
| Fuel (L)      | Positive decimal                       | Liters  | Heuristic from distance / efficiency x traffic; <b>not</b> a calibrated fleet figure. |
| Route         | Ordered list of strings                | —       | <b>Origin</b> → <b>waypoints</b> → <b>destination</b> path labels.                    |

Tab 3 - Warehouse

| Metric                   | Range / values                                 | Unit | Interpretation                                           |
|--------------------------|------------------------------------------------|------|----------------------------------------------------------|
| Estimated Walk Reduction | ≤ 35 (cap), grows with task count in prototype | %    | Illustrative <b>pick-path</b> savings vs naive ordering. |
| High-demand zones        | List of aisle labels                           | —    | Zones where <b>demand_score</b> ≥ 0.8 on demo tasks.     |
| Optimized sequence       | Ordered steps                                  | —    | Pick tour order for the demo SKU list.                   |

Tab 4 - Maintenance

| Metric     | Range / values                      | Unit  | Interpretation                                                                                                   |
|------------|-------------------------------------|-------|------------------------------------------------------------------------------------------------------------------|
| Risk       | Low · Medium · High                 | —     | From <b>risk_score</b> thresholds: ≥ 0.7 High, ≥ 0.45 Medium, else Low (demo inputs fixed in orchestrator).      |
| Risk Score | 0 - ~1+ before clamping-style bands | Score | Weighted blend of <b>temperature</b> , <b>vibration</b> , <b>mileage</b> , <b>DTC count</b> (prototype weights). |
| Action     | Text                                | —     | Recommended next step for that score band.                                                                       |

Tab 5 - Fraud

| Metric            | Range / values               | Unit        | Interpretation                                                                                                                                |
|-------------------|------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Fraud Probability | 0.000 - 0.990                | Probability | Blended score from <b>claim size</b> , <b>claim frequency</b> , <b>missing docs</b> , <b>account age</b> (fixed demo claim in orchestration). |
| Decision          | auto_process · manual_review | —           | <b>manual_review</b> if fraud probability ≥ 0.6.                                                                                              |

Tab 6 - Customer Service (chatbot)

| Metric                | Range / values | Unit | Interpretation                                                               |
|-----------------------|----------------|------|------------------------------------------------------------------------------|
| Answer                | Text           | —    | Retrieval-grounded response from local offline RAG index.                    |
| Escalate to human     | Yes / No       | —    | Triggered by confidence threshold or complaint signal.                       |
| Customer Service Chat | Interactive    | —    | Chat-style demo window allows repeated user questions and assistant replies. |

Tab 7 - Pricing

| Metric            | Range / values             | Unit            | Interpretation                                                   |
|-------------------|----------------------------|-----------------|------------------------------------------------------------------|
| Recommended Price | ≥ 3.5 after rounding       | Currency (demo) | Final <b>dynamic price</b> .                                     |
| Explainability    | Layman + technical details | —               | Presented as <b>Inputs / Components / Formula / Thresholds</b> . |

Tab 8 - Last-Mile Failure

| Metric              | Range / values                            | Unit        | Interpretation                                                      |
|---------------------|-------------------------------------------|-------------|---------------------------------------------------------------------|
| Failure Probability | 0.000 - 0.990                             | Probability | Same model as dashboard <b>Failure Probability</b> (see section 5). |
| Action              | normal_dispatch · confirm_delivery_window | —           | <b>confirm_delivery_window</b> if failure probability ≥ 0.5.        |

Tab 9 - Carbon

| Metric                   | Range / values                        | Unit | Interpretation                                                                                                            |
|--------------------------|---------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------|
| Estimated Emissions (kg) | ≥ 0                                   | kg   | Distance × per- <b>vehicle_type</b> factor (EV uses <b>grid_carbon_intensity</b> scaling).                                |
| Saving vs Diesel         | 0 - 100 (percentage points)           | %    | Improvement vs a <b>diesel baseline</b> for the same distance.                                                            |
| Recommendation           | switch_to_ev · optimize_consolidation | —    | <b>switch_to_ev</b> when urgency is <b>standard</b> and distance ≤ 120 km in the prototype; otherwise consolidation hint. |

Tab 10 - Multi-Agent (summary)

Repeats **Dispatch Buffer**, **Final Price**, **Failure Probability**, **Agent Notes**, and optionally **Replay Trace** — same interpretation as in sections 5–6.

## 8) Recommended Usage Patterns

- **A/B testing:** Fix all inputs except one (e.g. only **Weather**) and compare **ETA**, **Price**, **Failure Probability**, **CO<sub>2</sub>**.
- **Reliability:** Sweep **Tool Retries** (0 → 3) and observe errors/timeouts in **raw JSON**.
- **Latency stress:** Lower **Tool Timeout** toward **1.0 s** and combine with high **Traffic / storm**.
- **Explainability:** **Mode** = **replay** + **Show raw JSON** to align numbers with **trace** steps.

## 9) Troubleshooting

| Issue | What to check |
|-------|---------------|
|       |               |

|                                    |                                                                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------|
| Connection errors                  | <b>FastAPI Base URL</b> matches <code>uvicorn</code> ; firewall/host binding.                 |
| HTTP errors                        | Enable <b>raw JSON</b> ; read response body from API.                                         |
| Empty replay trace                 | <b>Mode</b> must be <b>replay</b> ; backend must expose <code>/agents/control/replay</code> . |
| Metrics unchanged after code edits | Restart <b>backend</b> (or confirm <code>--reload</code> picked up files).                    |
| Scenario uses fallback estimates   | <code>GOOGLE_MAPS_API_KEY</code> may be missing; app falls back to heuristic mode by design.  |

For onboarding, **change one input at a time** and compare the **plan** metrics (section 5) before diving into every challenge tab.

## 10) Demo Script (3 minutes)

Use this as a standard talk track for judges/stakeholders.

### 0:00 - 0:30 | Set context

1. Open the simulator and point to:
2. **Mode** selector
3. **View** selector
4. **Agents/Tools** overview
5. Say:
6. "This is a multi-agent logistics control tower prototype covering demand, route, warehouse, maintenance, fraud, customer service, pricing, failure risk, and carbon optimization."

### 0:30 - 1:20 | Create realistic scenario

1. In **Real-life Scenario Builder**:
2. Set **Collection Point (Start)** and **Delivery Point (Destination)**
3. Select **Scenario Vehicle Type** ( `van` or `motorcycle` )
4. Click **Create Scenario (Google Maps + Weather Forecast)**
5. Highlight generated summary:
6. `distance / travel time / traffic / weather`
7. fallback note if key is not configured
8. Say:
9. "We auto-generate route and context signals, then estimate Demand Index and Fleet Health Index with explainability."

### 1:20 - 2:20 | Run orchestration and show business outputs

1. Click **Run Simulation** (prefer `replay` mode for explainability demo).
2. In top metrics, call out:
3. **ETA, Price Quote, Failure Probability, CO2**
4. **KPI deltas** vs baseline
5. Open tabs quickly:
6. **Route** (distance alignment and ETA)
7. **Fraud** (probability + decision)
8. **Carbon** (saving vs diesel)
9. Say:
10. "Each challenge now provides plain-language explainability and technical breakdowns."

### 2:20 - 2:50 | Customer service AI chat

1. Go to **Challenge 6: RAG Customer Service**.
2. Ask one live question in the chat box (e.g., delivery ETA or claim process).
3. Say:
4. "The support assistant returns retrieval-grounded responses and indicates escalation when confidence is low."

### 2:50 - 3:00 | Close

1. Show **Eval Metrics** panel (if enabled in sidebar).
2. Say:
3. "The solution combines operational KPIs, agent orchestration, and transparent explainability with an evaluation endpoint for auditability."